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HOME / SERVICES / POWER ELECTRONICS PROTOTYPING

SERVICE 02 / POWER CONVERSION

Power electronics *on the bench.*

Converter prototypes become easier to improve when bring-up, switching behavior and measurements are treated as one engineering problem.

CONVERTERS   PCB PROTOTYPING   BRING-UP   VALIDATION

01 / PROBLEMS THIS ADDRESSES

When the waveform *raises the question.*

Component choices, transformer behavior, snubbers and layout decisions often interact. The useful next step is a measured one.

Typical starting points

A low-power converter that needs controlled bring-up, incomplete bench notes, unexplained switching behavior or a design decision that cannot be separated from the measurement setup.

02 / TECHNICAL CAPABILITIES

Use the bench to *guide the revision.*

POWER STAGE

Architecture and components

Converter topology, magnetics, component selection, snubbers and prototype schematics within the agreed scope.

BRING-UP

Measured behavior

Oscilloscope observations, differential probing, electronic-load tests, waveform review and stated test conditions.

03 / ENGAGEMENT WORKFLOW

From power stage *to evidence.*

01

Define the test boundary

Voltage, load, access and safety conditions.

02

Review the design

Topology, components, magnetics and layout context.

03

Bring up carefully

Use controlled power, probing and current limits.

04

Compare measurements

Review switching behavior against the design question.

05

Record the next revision

Capture observations, boundaries and priorities.

PUBLIC TECHNICAL PROTOTYPE

5V / 1A SMPS

Offline flyback prototype with a custom hand-wound RM10/I transformer, TNY285 controller, schematics, BOMs, waveforms and bench validation notes.

VIEW PROJECT   VIEW REPOSITORY ↗

04 / REPRESENTATIVE EVIDENCE

A documented *power-stage review.*

This is public technical work, not a completed client contract or certified product. The repository is the source for the published description and evidence boundary.

05 / DELIVERABLES

Evidence for the *next revision.*

- Converter architecture or component-selection notes
- Prototype schematic, BOM or PCB observations
- Bring-up checklist and measured waveforms

- Electronic-load or component-test observations
- Validation notes with prioritized next steps

> **Measurements need context.**
>
> Results are reported with the hardware, instrumentation and test conditions that shaped them.

START A TECHNICAL DISCUSSION

Bring the *prototype.*

Describe the converter, current stage, measured behavior and the design decision you need to make. Secure file exchange can follow the initial discussion.

START A TECHNICAL DISCUSSION →

Or write directly: info@herdersystemen.com

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